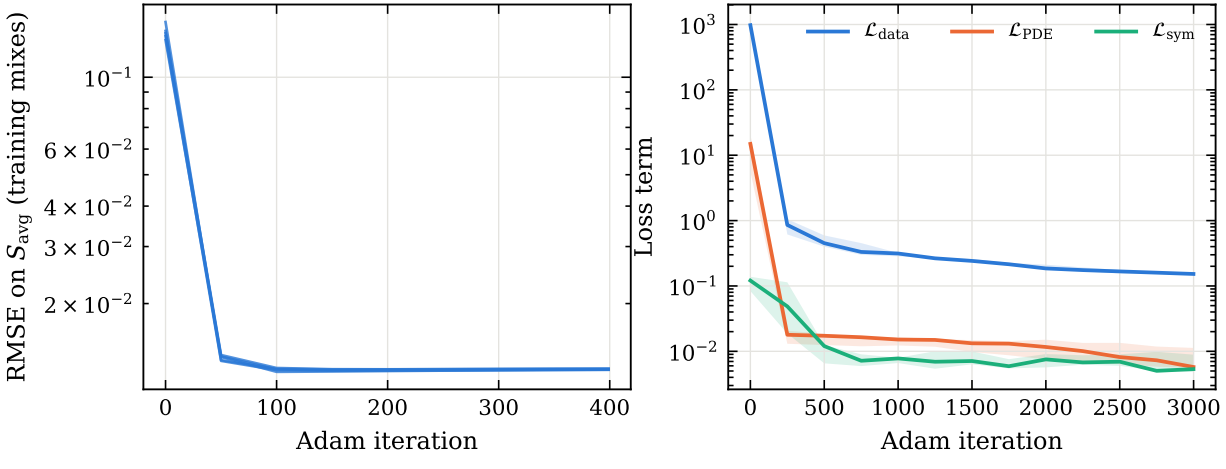
(a)
Physics-embedded network (5 seeds)
RMSE on S_{avg} (training mixes)
10^{-1}
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
0
100
200
300
400
Adam iteration
(b)
PINN (3 seeds)
Loss term
$\mathcal{L}_{data}$
$\mathcal{L}_{PDE}$
$\mathcal{L}_{sym}$
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
0
500
1000
1500
2000
2500
3000
Adam iteration